

Evidence Quality

Workbook 2 - Apply - Focused practice

Directions

Treat each item as close-choice practice. Preserve scope, certainty, and the exact job of each piece of evidence.

Exercise 1

A company reports that customer complaints fell 22 percent during a six-week chatbot pilot. Overall call volume also fell 19 percent during the same period.

Task: What is the best evaluation?

- A. The complaint decline proves the chatbot improved service.
- B. The complaint decline is irrelevant.
- C. The decline is relevant, but lower call volume means more evidence is needed to know whether the chatbot caused the change.
- D. A six-week pilot can never provide useful evidence.

Exercise 2

A city tests protected bike lanes on one downtown street for three weekends and records fewer vehicle-bicycle conflicts.

Task: Which conclusion is best supported?

- A. Protected bike lanes eliminate bicycle crashes everywhere.
- B. The pilot suggests the design may reduce conflicts at that location, but broader evidence is needed before a citywide claim.
- C. The test proves drivers prefer bicycles.
- D. No useful conclusion can be drawn from the pilot.

Exercise 3

A school wants to know whether a later start time improves attendance. It compares attendance this semester with last semester, but this semester also introduced free morning buses.

Task: What additional evidence would most improve sufficiency?

- A. Attendance patterns among similar students with and without access to the new bus service.
- B. The school mascot.
- C. A list of teacher birthdays.
- D. The price of lunch.

Exercise 4

A survey of 18 volunteers finds that 16 prefer a new park design. The city has 24,000 residents.

Task: Why is the survey not sufficient for a claim that "residents overwhelmingly support the redesign"?

- A. Eighteen is an even number.
- B. The sample is very small and may not represent the wider population.
- C. Volunteer surveys are never relevant.
- D. Preferences cannot be surveyed.

Answer reasoning

Check your reasoning, not only your answer.

For every question, compare your reasoning with the explanation. If you missed the item, identify the exact step where your reasoning changed the source or used evidence for the wrong job.

Exercise 1

Best answer: C. The decline is relevant, but lower call volume means more evidence is needed to know whether the chatbot caused the change.

Why: The evidence relates to the claim, but a competing change could explain part of the decline.

Exercise 2

Best answer: B. The pilot suggests the design may reduce conflicts at that location, but broader evidence is needed before a citywide claim.

Why: It preserves both the useful result and the limited scope.

Exercise 3

Best answer: A. Attendance patterns among similar students with and without access to the new bus service.

Why: It helps separate the effect of start time from the new transportation support.

Exercise 4

Best answer: B. The sample is very small and may not represent the wider population.

Why: The main problem is representativeness and scope, not whether surveys can ever provide evidence.

Mistake check

For every miss, label the reason: wrong claim, evidence mismatch, too broad or too certain, reasoning gap, source or format confusion, or rushed reading. Then write one sentence explaining what you will check next time.